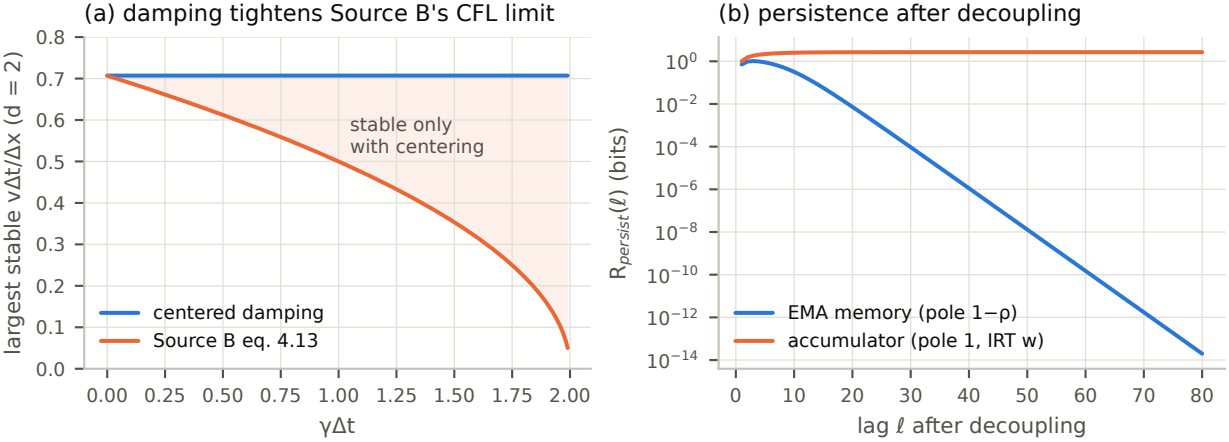
(a) damping tightens Source B's CFL limit
largest stable vΔt/Δx (d = 2)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
stable only
with centering
centered damping
Source B eq. 4.13
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
γΔt
(b) persistence after decoupling
$R_{persist}(\ell)$ (bits)
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
EMA memory (pole 1−ρ)
accumulator (pole 1, IRT w)
0
10
20
30
40
50
60
70
80
lag ℓ after decoupling